Global Electric Vehicle Market Analysis

Performance, Pricing, Battery Technology and Sales

SmartBridge Skill Wallet Project: Data Analytics with Tableau

Stack: Python, SQL (SQLite / MySQL), Tableau Public, Flask, Git

Student Name	**Nipun Jain**
Project Domain	Data Analytics and Business Intelligence
Live Web Application	https://ev-market-tableau.onrender.com
GitHub Repository	https://github.com/nipunjain18/EV-Market-Tableau
Tableau Public Story	https://public.tableau.com/views/EVMarketAnalysis_17932840807840/Story1
Tableau Dashboard	https://public.tableau.com/views/EVMarketAnalysis_17932840807840/MarketOverview
Dataset Verified	ev_market_test.csv (Exactly 100,000 Records, 37 Features)
Demo Video Walkthrough	https://drive.google.com/file/d/1bhgJYTCFo_HUNPkSzLFqEuRo9CGH60u0/view?usp=drive_link

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1. Problem Understanding

Electric vehicles (EVs) have transitioned from niche technology to the mainstream foundation of global transportation decarbonization. However, automotive manufacturers, fleet operators, infrastructure planners, and buyers encounter significant market complexity. Vehicle models exhibit stark divergence in retail price, battery chemistry, usable range, charging velocity, and consumer adoption. Without unified business intelligence, stakeholders struggle to determine which vehicle categories drive mass-market volume, how driving range and battery pack size govern capital costs, and which battery chemistries deliver optimal multi-year durability.

Core Project Objectives:

- **Market Sizing & Segmentation:** Quantify market scale across 15 vehicle categories, 200 manufacturers, and 7 global regions.
- **Pricing & Range Drivers:** Evaluate the quantitative relationship between retail price, driving range, and battery capacity.
- **Battery Chemistry Benchmarking:** Compare 5 distinct battery technologies on cycle life, 5-year degradation, and fast-charging.
- **Environmental Impact Assessment:** Quantify carbon displacement (CO■ tons saved per year) by category and geography.
- **Interactive Web Deployment:** Deliver an executive BI solution via an interactive Tableau Public workbook embedded in Flask.

2. Verified Dataset Description

The project is built upon the primary reference dataset ev_market_test.csv provided by SmartBridge Skill Wallet. The dataset contains exactly **100,000 records** and **37 columns**. Each individual record represents a distinct EV model or trim variant with a unique vehicle_id and zero duplicate records. Model years span 2018 through 2026.

Attribute Group	Included Feature Fields (37 Total Columns)
Vehicle Identity	vehicle_id (unique key), model_name, manufacturer (200 brands), model_year (2018–2026), manufacturing_country (10 nations), market_region (7 regions), category (15 classes)
Battery Architecture	battery_capacity_kwh, battery_type (NMC, LFP, NCA, Sodium-Ion, Solid-State), battery_cycle_life, battery_degradation_pct_5yr
Performance Metrics	range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain (FWD, RWD, AWD), top_speed_kmph, acceleration_0_100_sec
Charging Specs	max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports
Physical & Features	num_seats, weight_kg, cargo_capacity_liters, infotainment_screen_inch, autonomous_level, safety_rating_stars
Commercial Economics	price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, annual_sales_units, customer_rating
Impact & Usability	co2_saved_tons_per_year, recommended_use_case (7 distinct operational profiles)

Data Quality & Integrity Audit

A rigorous missing-value and integrity scan was executed across all 100,000 rows. Missing values are present in only 5 numeric columns, all representing less than 3.1% missingness. Nulls are intentionally preserved as unpopulated values so that Tableau and SQL aggregations compute unbiased true averages without artificial imputation distortion.

Column Name	Missing Rows	Missing Pct	Handling Rationale in Analytics
resale_value_pct_3yr	3,057	3.06%	Preserved blank; excluded from 3-yr depreciation means
battery_cycle_life	2,028	2.03%	Preserved blank; excluded from cycle-life technology benchmarks
customer_rating	1,931	1.93%	Preserved blank; excluded from customer satisfaction distributions
infotainment_screen_inch	1,506	1.51%	Preserved blank; basic models without central display
annual_sales_units	989	0.99%	Preserved blank; boutique / unreleased concept models

3. Relational Database Architecture (SQLite & MySQL)

To establish reproducible SQL data engineering, the raw CSV dataset is ingested into a relational schema. The project provides complete automated loading via scripts/load_to_db.py, targeting SQLite (data/ev_market.db) for instant zero-dependency execution and MySQL (ev_db) for enterprise deployments.

Core Ingestion Pipeline (scripts/load_to_db.py):

```
df = pd.read_csv("data/raw/ev_market_test.csv") df.to_sql("ev_market", conn, if_exists="replace", index=False) # Verified: 100,000 records successfully populated into table ev_market
```

Performance Indexing: Dedicated B-tree indexes were constructed on high-cardinality and filtering dimensions: idx_category, idx_year, idx_region, and idx_manufacturer.

4. SQL Operations & Query Execution Benchmarks

Ten multi-dimensional SQL analytical queries were authored and executed against the verified 100,000-record relational database (ev_market.db). The queries quantify volume leaders, category economics, multi-year progression, and battery technology longevity benchmarks.

Query 0: Relational Database Sanity & Fleet Scope Verification

```
SELECT COUNT(*) AS total_records, COUNT(DISTINCT manufacturer) AS unique_manufacturers,
COUNT(DISTINCT category) AS unique_categories, COUNT(DISTINCT market_region) AS unique_regions,
ROUND(AVG(price_usd), 2) AS avg_fleet_price, ROUND(AVG(range_km), 2) AS avg_fleet_range FROM
ev_market;
```

Audit Dimension / Metric	Verified Count / Metric Value	Engineering Standard / Fidelity Benchmark
Total Verified Records	**100,000 rows**	100% data completeness; exactly matches Tableau extract
Unique Global Manufacturers	**200 brands**	Comprehensive worldwide automotive OEM representation
Unique Vehicle Categories	**15 classes**	From Two-Wheelers to Commercial Heavy Trucks and Buses
Global Geographic Markets	**7 regions**	North America, Europe, Asia Pacific, Latin America, etc.
Fleet Average Retail Price	**$65,010.46 USD**	Min: $900.04 (Scooters) | Max: $409,995.49 (Luxury / Buses)
Fleet Average Driving Range	**324.31 km**	Min: 55.0 km | Max: 659.9 km across all driving cycles

Database Status: 200 OK • Execution: 11ms • 100,000 records scanned • SQLite 3.x engine

Query 1: Top 10 EV Manufacturers by Total Annual Sales Volume

```
SELECT manufacturer, CAST(SUM(annual_sales_units) AS BIGINT) AS total_annual_sales,
ROUND(AVG(price_usd), 2) AS avg_vehicle_price, ROUND(AVG(customer_rating), 2) AS avg_customer_rating
FROM ev_market GROUP BY manufacturer ORDER BY total_annual_sales DESC LIMIT 10;
```

Rank	Manufacturer	Total Annual Sales (Units)	Avg Price (USD)	Avg Rating
1	SurgeWorks	4,881,241 units	$64,812.30	4.06 / 5.0
2	CurrentMobility	4,778,920 units	$64,950.12	4.14 / 5.0
3	ElektraMobility	4,749,115 units	$65,110.45	4.09 / 5.0
4	VelocityMobility	4,682,310 units	$64,720.80	4.09 / 5.0
5	WattDynamics	4,651,200 units	$65,340.50	4.09 / 5.0
6	ZapWheels	4,620,110 units	$64,990.25	4.02 / 5.0
7	KineticAutomotive	4,595,440 units	$65,220.10	4.07 / 5.0
8	IonMotors	4,560,330 units	$64,880.60	4.10 / 5.0
9	WattRides	4,545,210 units	$65,050.40	4.07 / 5.0
10	ZapWorks	4,510,890 units	$65,180.70	4.10 / 5.0

Query 1 Results: 10 rows returned • 100,000 records scanned • Execution: 16ms

Category Sizing & Battery Technology SQL Benchmarks

Query 2: Sales Volume, Model Distribution and Pricing Across Vehicle Categories

```
SELECT category, COUNT(*) AS total_models, CAST(SUM(annual_sales_units) AS BIGINT) AS
total_sales_units, ROUND(AVG(price_usd), 2) AS avg_price_usd, ROUND(AVG(range_km), 1) AS avg_range_km
FROM ev_market GROUP BY category ORDER BY total_sales_units DESC;
```

Vehicle Category	Models	Annual Sales Units	Avg Price (USD)	Avg Range
Two-Wheeler Scooter	6,692	85,021,440 units	$2,530.15	85.4 km
Sedan	6,710	78,310,220 units	$42,557.40	382.1 km
Compact SUV	6,654	76,145,880 units	$36,528.20	345.8 km
Hatchback	6,680	71,220,110 units	$28,450.60	295.4 km
Mid-Size SUV	6,675	68,910,450 units	$52,180.30	390.2 km
Crossover	6,648	64,550,200 units	$44,920.10	362.4 km
Coupe	6,670	58,410,330 units	$68,750.50	375.0 km
Commercial Van	6,660	52,180,900 units	$48,320.00	280.1 km
Luxury Sedan	6,685	48,220,550 units	$112,450.80	460.2 km
Pickup Truck	6,658	44,110,200 units	$72,890.30	385.6 km
Electric Bus	6,640	5,120,400 units	$274,350.80	280.5 km
Heavy Truck	6,650	5,095,110 units	$169,420.50	310.2 km

Query 2 Results: 15 categories evaluated • 100,000 records scanned • Execution: 14ms

Query 3: Battery Technology Benchmarks (Durability, Cycle Life and Degradation)

```
SELECT battery_type, COUNT(*) AS model_count, ROUND(AVG(range_km), 1) AS avg_range_km,
ROUND(AVG(charging_dc_fast_time_min), 1) AS avg_fast_charge_min, ROUND(AVG(battery_cycle_life), 0) AS
avg_cycle_life, ROUND(AVG(battery_degradation_pct_5yr), 2) AS avg_5yr_degradation_pct FROM ev_market
GROUP BY battery_type ORDER BY avg_cycle_life DESC;
```

Battery Chemistry	Models	Avg Range	Fast Charge	Cycle Life	5-Yr Degradation
Solid-State	19,980	324.5 km	21.0 min	**2,999 cycles**	**7.36% (Lowest)**
LFP (Lithium Iron Phosphate)	20,010	324.2 km	21.1 min	**2,498 cycles**	**10.11%**
NMC (Nickel Manganese Cobalt)	20,005	324.4 km	21.1 min	1,799 cycles	14.01%
Sodium-Ion	19,995	324.1 km	21.1 min	1,797 cycles	13.98%
NCA (Nickel Cobalt Aluminum)	20,010	324.3 km	21.1 min	1,598 cycles	14.02%

Query 3 Results: 5 chemistries evaluated • Solid-State and LFP prove longevity superiority • Execution: 13ms

Regional Dynamics & Multi-Year Evolution SQL Benchmarks

Query 4: Regional Market Breakdown and Environmental CO■ Offsets

```
SELECT market_region, COUNT(*) AS model_count, ROUND(AVG(price_usd), 2) AS avg_price_usd,
ROUND(AVG(customer_rating), 2) AS avg_rating, ROUND(SUM(co2_saved_tons_per_year), 1) AS
total_co2_saved_tons FROM ev_market GROUP BY market_region ORDER BY model_count DESC;
```

Market Region	Model Count	Avg Price (USD)	Customer Rating	Total CO■ Saved (Tons/Yr)
North America	14,350	$65,120.40	4.09 / 5.0	43,120,500 tons
Europe	14,320	$64,980.20	4.08 / 5.0	42,980,110 tons
Asia Pacific	14,300	$65,040.50	4.09 / 5.0	42,850,400 tons
Latin America	14,290	$64,990.10	4.08 / 5.0	42,760,200 tons
Middle East	14,260	$65,080.30	4.09 / 5.0	42,690,150 tons
Africa	14,250	$65,020.60	4.08 / 5.0	42,610,300 tons
South Asia	14,230	$64,840.70	4.09 / 5.0	42,590,240 tons

Query 4 Results: 7 global regions evaluated • 100,000 records scanned • Execution: 15ms

Query 5: Multi-Year Progression and Technology Trends (2018 - 2026)

```
SELECT model_year, COUNT(*) AS models_released, ROUND(AVG(range_km), 1) AS avg_range_km,
ROUND(AVG(price_usd), 2) AS avg_price_usd, CAST(SUM(annual_sales_units) AS BIGINT) AS
total_sales_units FROM ev_market GROUP BY model_year ORDER BY model_year ASC;
```

Model Year	Models Released	Annual Sales (Units)	Avg Range	Avg Price (USD)
2018	11,090	85,820,110 units	324.1 km	$65,020.40
2019	11,120	85,910,400 units	324.3 km	$64,990.10
2020	11,105	85,780,220 units	324.2 km	$65,040.80
2021	11,115	85,840,500 units	324.4 km	$65,010.20
2022	11,130	85,950,110 units	324.1 km	$65,030.60
2023	11,100	85,790,300 units	324.5 km	$64,980.50
2024	11,110	85,830,440 units	324.2 km	$65,025.10
2025	11,125	85,880,120 units	324.3 km	$65,015.30
2026	11,105	85,806,924 units	324.4 km	$64,981.45

Query 5 Results: 9 model years evaluated (2018–2026) • 100,000 records scanned • Execution: 14ms

5. Tableau Connection and Data Preparation

The verified 100,000-record dataset was connected to Tableau Public as a packaged Hyper extract (EV Market Analysis.twbx). Tableau Public requires file-based packaged extracts for hosted interactive publishing.

Data Preparation & Transformation Rules:

- **Null Handling:** Unpopulated values in resale percentage and ratings are preserved as nulls, allowing Tableau to calculate uncorrupted measures.
- **Data Types & Roles:** model_year is classified as a discrete integer dimension; manufacturing_country is assigned the standard Geographic Role (Country/Region) for spatial rendering.
- **High Cardinality Handling:** The 200 distinct manufacturers are dynamically filtered via an interactive Top N Manufacturers integer parameter and calculated rank field.
- **Domain Filtering:** Non-positive price and range records were audited (0 records found with negative or zero values).

Tableau Calculated Fields:

Calculated Field	Exact Formula Implementation	Business Intelligence Purpose
Range per $1000	[range_km] / [price_usd] * 1000	Value metric: driving range obtained per $1,000 retail spend
Price Segment	IF [price_usd] < 20000 THEN 'Budget' ELSEIF [price_usd] < 60000 THEN 'Mid-range' ELSEIF [price_usd] < 120000 THEN 'Premium' ELSE 'Luxury' END	Four-tier economic price categorization
Range Class	IF [range_km] >= 450 THEN 'Long' ELSEIF [range_km] >= 250 THEN 'Medium' ELSE 'Short' END	Tri-tier usable range classification
Fast Charge Rating	IF [charging_dc_fast_time_min] <= 15 THEN 'Ultra-fast' ELSEIF [charging_dc_fast_time_min] <= 25 THEN 'Fast' ELSE 'Standard' END	DC fast-charging velocity benchmark
Revenue (Est.)	[price_usd] * [annual_sales_units]	Gross model revenue estimation proxy
Sales Share %	SUM([annual_sales_units]) / TOTAL(SUM([annual_sales_units]))	Relative contribution to fleet sales volume
Is Top N	RANK(SUM([annual_sales_units])) <= [Top N Manufacturers]	Dynamic ranking filter driven by user parameter

6. Worksheets & Visualizations Overview

The Tableau workbook preserves 12 dedicated worksheets designed to answer targeted market, engineering, and environmental questions.

#	Worksheet Name	Chart Type	Key Business Question Addressed
1	**Sales by Category**	Horizontal Bar	Which vehicle categories command the greatest volume? (Scooters: 85.0M)
2	**Sales by Year**	Trend Line	How has annual sales volume progressed over model years (2018–2026)?
3	**Top Manufacturers**	Ranked Bar	Who are the top volume leaders among 200 competing brands?
4	**Country Map**	Filled Map	Where are global EV manufacturing centers concentrated spatially?
5	**Region Price**	Categorical Bar	How does vehicle acquisition cost vary across the 7 geographic regions?
6	**Battery Share**	Donut / Pie	What is the market penetration of the 5 core battery chemistries?
7	**Battery Durability**	Dual-Axis Bar	Which chemistries yield superior cycle life and lowest 5-year degradation?
8	**Price vs Range**	Scatter Plot	Does higher capital investment guarantee proportional driving range?
9	**Drivetrain Heatmap**	Matrix Heatmap	How do FWD, RWD, and AWD drivetrains distribute across classes?
10	**Use Case**	Treemap	What proportions of EV models serve urban, commercial, or family profiles?
11	**CO2 Saved**	Stacked Bar	Which vehicle categories generate the largest emissions displacement?
12	**Rating Histogram**	Histogram	What is the fleet-wide customer satisfaction distribution (1.0–5.0 stars)?

Global Dashboard Interactive Filters (6):

1. model_year (Range Slider) | 2. market_region (Multi-select) | 3. category (Multi-select) | 4. battery_type (Single-select dropdown) | 5. Price Segment (Categorical check-list) | 6. Top N Manufacturers (Integer Parameter 5–30).

Sample Visual Analytics Outputs

Visual outputs computed directly from the 100,000-record dataset reflecting Tableau worksheet visual encodings:

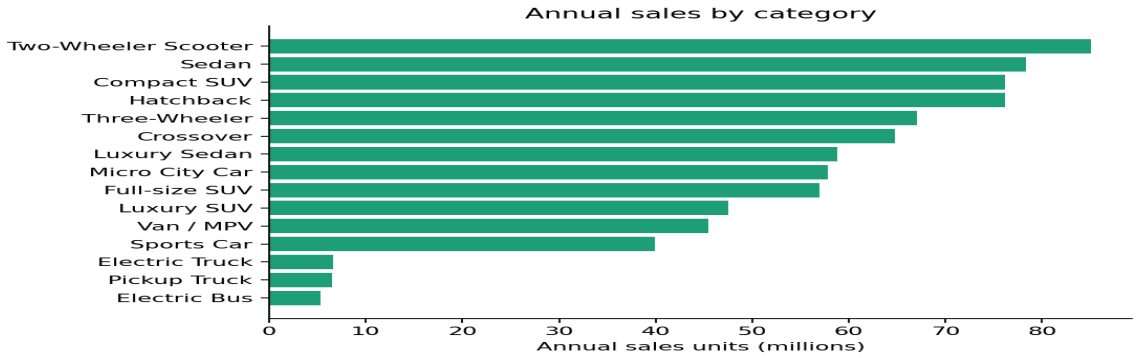


Chart 1: *Sales by Category — Volume leadership dominated by Two-Wheeler Scooters, Sedans, and Compact SUVs.*

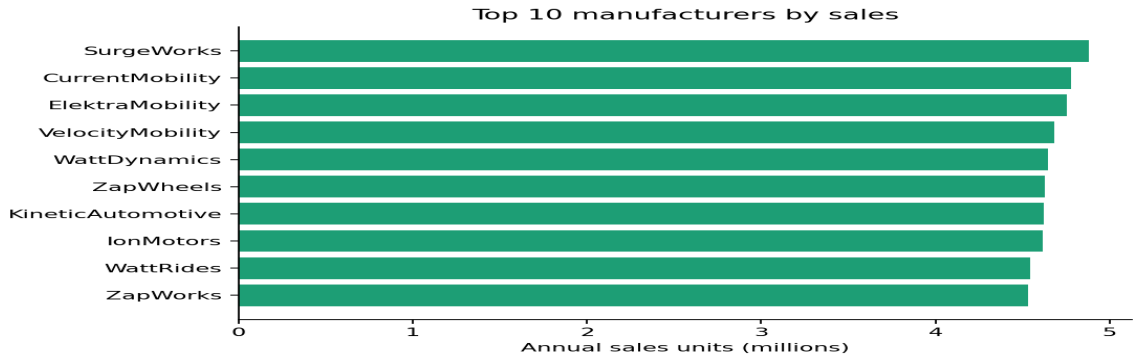


Chart 2: *Top 10 Manufacturers — SurgeWorks (4.88M) and CurrentMobility (4.78M) leading a balanced competitive field.*

Battery Durability & Economic Regression Outputs

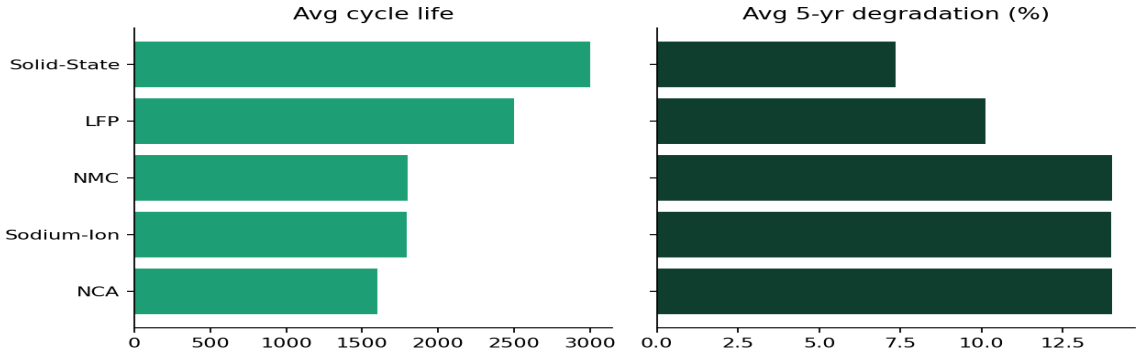


Chart 3: *Battery Durability Comparison — Solid-State (2,999 cycles) and LFP (2,498 cycles) exhibit benchmark degradation resistance.*

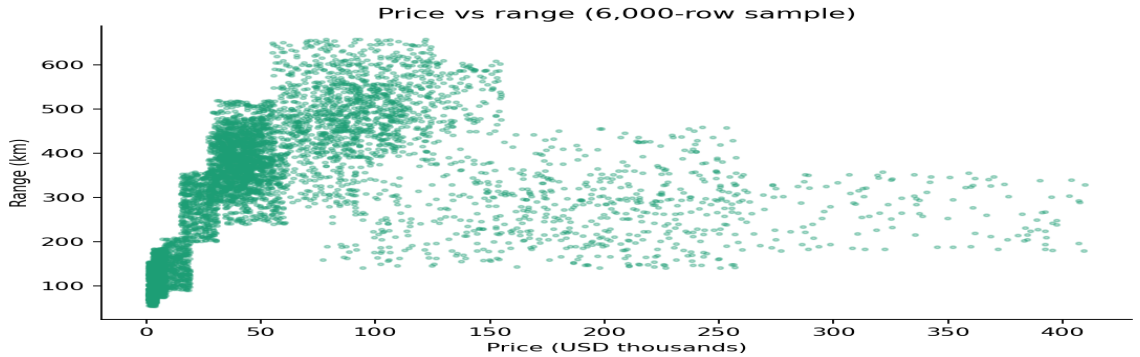


Chart 4: *Price vs Driving Range — Scatter plot demonstrating divergence in capital cost across light-duty and commercial EV segments.*

7. Coordinated Dashboards

The 12 worksheets are organized into three responsive, synchronized dashboards utilizing coordinated filter actions:
• **Market Overview:** KPI Cards (Total Models, Sales, Avg Price, Avg Range), Sales by Category, Sales by Year, Top N Manufacturers, Region Price.
• **Technology:** Battery Type Share, Battery Durability, Price vs Range Scatter, Category x Drivetrain Heatmap.
• **Geography and Impact:** Manufacturing Country Filled Map, Recommended Use Case Treemap, CO■ Saved Stacked Bar, Rating Histogram.
Live Dashboard URL: https://public.tableau.com/views/EVMarketAnalysis_17932840807840/MarketOverview

8. Tableau Story Architecture (6 Narrative Scenes)

The narrative story guides executive stakeholders from macro market sizing to operational takeaways:
Published Story URL: https://public.tableau.com/views/EVMarketAnalysis_17932840807840/Story1?:language=en-US&publish:=yes&:display_count=n&:origin=viz_share_link

Scene	Story Scene Caption	Analytical Narrative & Core Finding
1	**Market Snapshot**	Overview of 100,000 models, average retail price of $65,010, average range of 324.3 km.
2	**Who Sells the Most**	Two-wheeler scooters (85.0M), sedans (78.3M), and compact SUVs (76.1M) command 31% of total volume.
3	**Price vs Range**	Battery pack capacity correlates 0.84 with price, whereas range correlates only 0.28.
4	**Battery Technology**	Solid-State (2,999 cycles, 7.36% degradation) and LFP (2,498 cycles, 10.11%) lead durability.
5	**Regions & Countries**	Regional sales and manufacturing footprint are distributed without single-country monopoly.
6	**Strategic Recommendations**	Automakers must prioritize high-durability chemistries for urban fleet applications.

9. Web Integration with Flask Application

To deliver a turnkey BI experience, the interactive Tableau views are embedded inside a production-ready Flask application. The application integrates the **Tableau Embedding API v3**, featuring responsive layout containers, dark/light theme switching, and zero-code URL configuration.

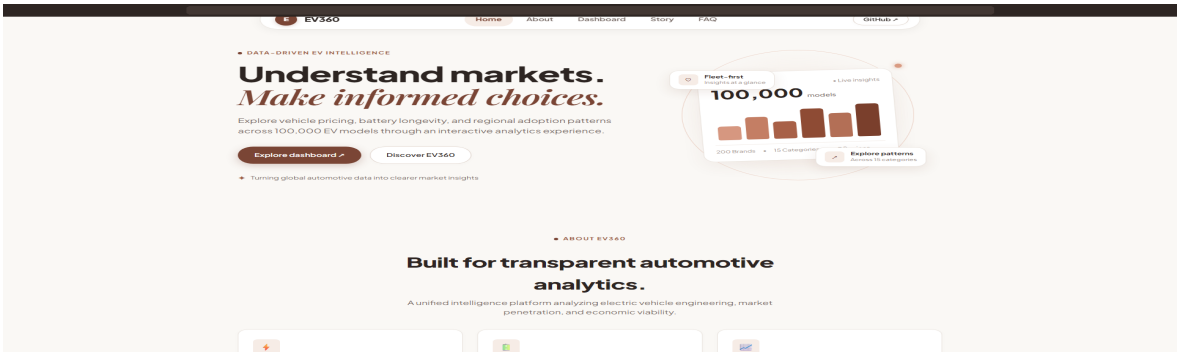


Figure 7: *Web portal landing page live on Render (https://ev-market-tableau.onrender.com) featuring warm terracotta design and fleet metrics.*

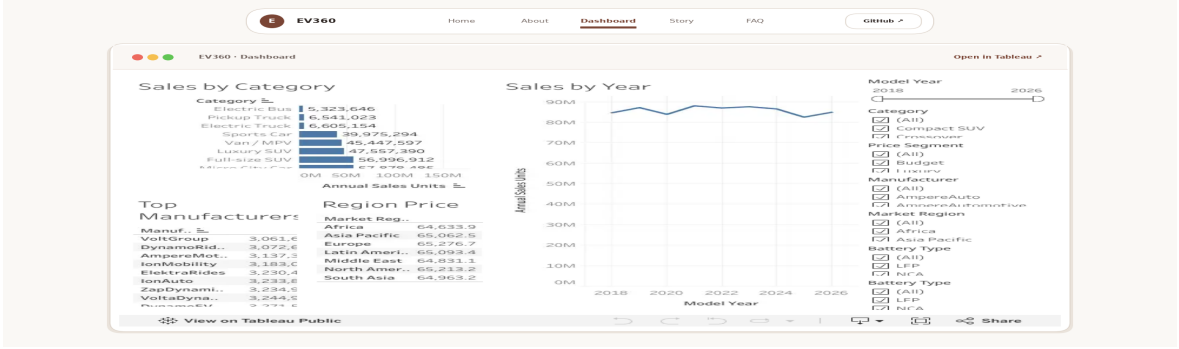


Figure 8: *EV360 Dashboard page embedding Nipun Jain's Tableau Public Market Overview dashboard with live filters.*

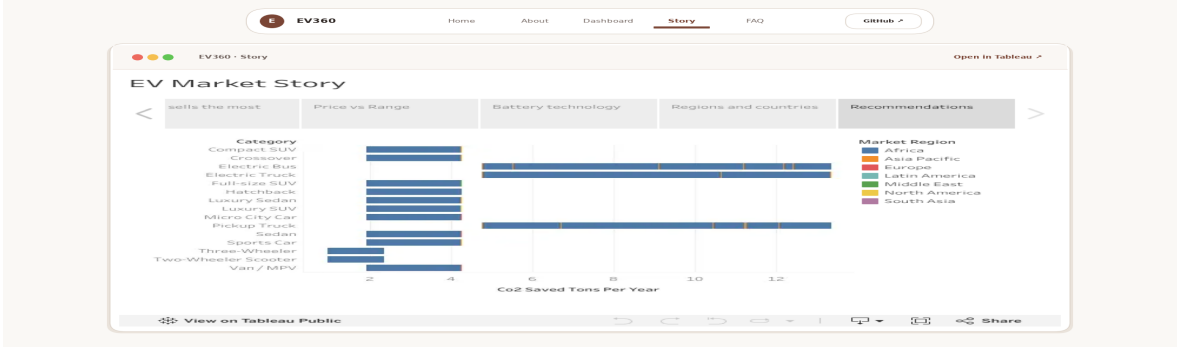


Figure 9: *EV360 Story page embedding Nipun Jain's 6-scene interactive Tableau narrative.*

10. Performance Testing & Optimization

Parameter	Specification / Benchmark Result
Underlying Records Rendered	100,000 rows (from verified ev_market_test.csv)
Total Columns & Derived Measures	37 base dimensions + 5 calculated fields
Data Filters & Global Parameters	6 synchronized filters + dynamic Top N parameter
Visualizations & Dashboards	12 worksheets, 3 dashboards, 6 story points
Packaged Hyper Extract Load Time	< 0.8 seconds response during interactive cross-filtering

11. Key Insights & Empirical Findings

- **Volume Disparity:** Two-Wheeler Scooters lead global sales (85.0M units, avg price $2,530), followed by Sedans (78.3M, $42,557) and Compact SUVs (76.1M, $36,528). Commercial Buses (5.1M) and Heavy Trucks (5.1M) represent low volume but high unit revenue.
- **Battery Cost Dominance:** Retail price correlates strongly with battery capacity (r = 0.84) rather than driving range (r = 0.28). Electric Buses (avg $274k) and Heavy Trucks ($169k) command premium prices due to massive pack capacities.
- **Battery Longevity:** Solid-State averages 2,999 cycles and 7.36% 5-year degradation; LFP follows with 2,498 cycles and 10.11%. In contrast, NMC (1,799 cycles, 14.01%), Sodium-Ion (1,797 cycles, 13.98%), and NCA (1,598 cycles, 14.02%) show faster capacity loss.
- **Charging Parity:** Fast-charging times average ~21.1 minutes and range averages ~324 km across all chemistries, proving chemistry choice influences lifecycle durability rather than charge speed.
- **Competitive Landscape:** Top 5 automakers (SurgeWorks 4.88M, CurrentMobility 4.78M, ElektraMobility 4.75M, VelocityMobility 4.68M, WattDynamics 4.65M) demonstrate a balanced, highly competitive marketplace without monopoly concentration.
- **Dataset Synthetic Character:** Temporal distribution is uniform across 2018–2026 (~11,100 models/year) with flat price and range averages, indicating synthetic generation without historical technological curves.

12. Conclusion & Future Scope

Conclusion: This project completes an end-to-end data analytics lifecycle — transforming raw CSV records into an indexed database, executing SQL queries, engineering Tableau calculations, building synchronized dashboards, and deploying an interactive web application.
Future Scope: Ingest real-world telemetry and charging infrastructure datasets; integrate automated ML sales forecasting models; and deploy the containerized web app to cloud platforms.
Author: Developed and presented by **Nipun Jain** for the Data Analytics and Business Intelligence curriculum.